To verify that the program always does what it is supposed to do, suppose that the precondition $P(x,y)$ holds. That is, we suppose that the statement "$x=a$ and $y=b$" is true. This means that $x=a$ and $y=b$. The first step of the program, $temp:=x$, assigns the value of x to the variable temp, so after this step we Know that $x=a$, $temp=a$ and $y=b$. After the second step of the program, $x:=y$, we Know that $x=b$, $temp=a$ and $y=b$. Finally, after the third step, we Know that $x=b$, $temp=a$ and $y=a$. Consequently, after this program is run, the postcondition $Q(x,y)$ holds, that is, the statement "$x=b$ and $y=a$" is true.

1.4.3 Quantifiers

Quantification expresses the extent to which a predicate is true over a range of elements. In English, the words all, some, many, none and few are used in quantifications. We will focus on two types of quantification here: universal quantification, which tells us that a predicate is true for every element under consideration, and existential quantification, which tells us that there is one or more element under consideration for which the predicate is true. The area of logic that deals with predicates and quantifiers is called the **predicate calculus**.

The Universal Quantifier

Many mathematical statements assert that a property is true for all values of a variable in a particular domain, called the **domain of discourse** (or the **universe of discourse**), often just referred to as the **domain**. Such a statement is expressed using universal quantification. The universal quantification of $P(x)$ for a particular domain is the proposition that asserts that $P(x)$ is true for all values of x in this domain. Note that the domain specifies the possible values of the variable x. The meaning of the universal quantification of $P(x)$ changes when we change the domain. The domain must always be specified when a universal quantifier is used; without it, the universal quantification of a statement is not